

Where AI belongs in construction project delivery

Less paperwork, earlier warning, better projects.

Ryan Prentice

Senior Project Manager, commercial real estate · Boston

Construction is a game of risk

Every project brings together parties carrying different risks, working with incomplete information, and making decisions whose consequences may not be visible for months. The project manager's job is to manage those risks and the relationships that keep them managed. That is the work that decides whether a project succeeds. It is also the work AI has started to change. The largest contractors and their software platforms have moved, and the owner's side is only beginning to.

THREE THINGS HAPPENING RIGHT NOW IN AI

\$700B

in AI infrastructure planned by the largest tech companies in 2026 alone

1 billion

monthly users on a single AI product, less than four years after launch

62%

of organizations already experimenting with AI agents that complete multi-step work

Sources. Hyperscaler 2026 capital spending guidance as reported by Reuters and Moody's. ChatGPT usage, June 2026. McKinsey, The State of AI, 2025.

AI will change how projects are run. The teams that build it into their workflows will deliver more consistently, with fewer surprises, and carry more work without adding people.

Here is what I believe after fifteen years running projects as an owner's representative and three years learning to build with AI. By my estimate, roughly three quarters of a project manager's week goes to administrative load, and that ratio is backwards. AI belongs first in the work that prepares a decision. It can draft the status report, reconcile the cost report against the latest invoices, and review a change order or pay application against the contract before the project manager picks it up. That gives the project manager back the time for judgment, relationships, and action.

That is the near-term value, and it is real. However, it is not the destination. Once AI is doing the first pass on the documents where risk hides, the schedule, the budget, the change orders, the submittals, the open decisions, it can start reading across all of them at once. From there, a synthesis layer can connect a pending decision to the milestone it threatens and flag it while there is still time to act. Project management is risk management, and that is ultimately what AI will be used for in construction.

Every hour AI takes off the administrative load is capacity. Every error a review catches before payment is money. Every risk surfaced before the milestone is time.

On a team of twenty project managers, moving the administrative load from three quarters of the week to half gives back roughly five people's worth of capacity without hiring anyone. At a firm with two hundred project managers, the same math is fifty. Time is only the first return, and page 07 sets out the full business case.

The path is deliberate. Automate the repeatable work first, build targeted reviews next, and connect them over time, with the project manager making every decision along the way. The rest of this document lays out that path and three working tools that show it in practice. The paperwork is the easy part. Seeing the risk early is the prize.

Where AI belongs in a project manager's week

Administration is most of the job today. Meetings, email, reporting, documentation, and keeping the numbers current. None of it is optional. Managing a budget means reporting on it, and managing a schedule means tracking every decision that impacts it. The work only a project manager can do, managing risk, making decisions, and keeping the relationships that move the project, sits inside that load and competes with it for the same hours.

AI can take much of the administrative side, and it can start now. In the near term that moves the balance to something like half the week, and the hours go back to that work. Longer term, once AI carries more of the load around meetings and email and takes on the first review of documents, the number is closer to a quarter. Those are estimates from my own experience. So the first step for any team is a baseline. Track where project managers actually spend their week, then build or buy the tools that go after the biggest blocks of time.

On most project teams that shift has barely started. Much of the conversation about AI in delivery still happens above the project team, in leadership discussions about strategy and tools. When a tool does arrive, it often comes as a license and a login, with little direction on what it is for, because too often no one is accountable for bringing it into the workflows. The project managers I know are not resistant. They have not had the guidance or the time to work out where it fits. What they need is a clear answer about which work AI does, which work it helps with, and which work stays with them. Being precise about that is the difference between a tool that gets used and one that gets ignored.

Automate

AI does it, the PM reviews

Status updates and reporting

Meeting notes and follow-up

Task creation and tracking

Filing documents

Routine email drafts

Augment

AI drafts, the PM decides

Budget and cost tracking

Change orders and pay applications

Schedule analysis

Submittal and drawing review

Procurement and contract review

Human only

Judgment stays human

Client conversations

Final risk decisions

Negotiation

Site walks

Team leadership

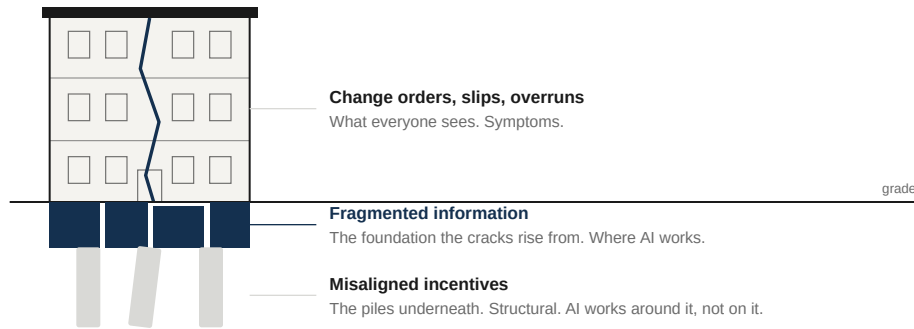
Risk management runs the full length. The project manager owns it end to end.

A change order shows all three in one workflow. AI can add the request to the change order log, save the package to the project folder, and create a task in the task management system. That is automation, work the tool does and the project manager checks. It can compare the markups with the contract, check the math, and flag scope that looks like rework. That is augmentation, a first review that makes the decision better informed. Establishing the facts, weighing the commercial position, and negotiating the outcome stay with the project manager.

The first gains come from automation, because the work is frequent and easy to check. The larger gains come from augmentation, because that is where the decisions are. In both, the measure is the effort it takes to reach an accepted result, and a tool that saves drafting time but creates more checking has only moved the burden. Time is the return a team feels first. Risk is where the larger return sits, and pointing AI at it starts with understanding where risk comes from.

The cracks show at the surface. The cause is in the foundation.

Ask anyone why projects go sideways and you get the same list. Scope creep, design changes, change orders, schedule slips, cost overruns. Those are symptoms. After years of watching projects drift and asking why, I have come to see the causes as more fundamental.



Underneath every symptom is risk. When a risk lands with a party that cannot manage it, it does not disappear. It comes back as contingency, disputes, and blown schedules, and the owner is usually the one left holding it. Protecting the owner from that is the job of the owner's project manager.

Risk lands in the wrong place because incentives do not align. The owner wants cost certainty, the contractor wants margin, and the architect protects the design. They are separate organizations with different obligations and authority, and the owner's project manager works across those boundaries, understanding where interests diverge and where a decision is being passed between parties. That misalignment is structural to how the industry is built, and no tool changes it.

Separate organizations also keep separate records. The schedule sits in the scheduler's software, the budget in a spreadsheet, the RFIs and submittals in the contractor's platform, and the decisions in email. The project manager is the connection between them, and the connections are where risk hides.

Take a budget that assumes a package includes a scope. The bid clarifications exclude it, and the drawings still require it. Each record looks routine on its own. Together they describe a scope gap that becomes a disputed change if nobody catches it. On a small project, the project manager catches it. On a large one, with hundreds of records moving every week, one person cannot hold all of it. That is not a failure of the person. It is the limit of how the work is set up and the tools the team has been given.

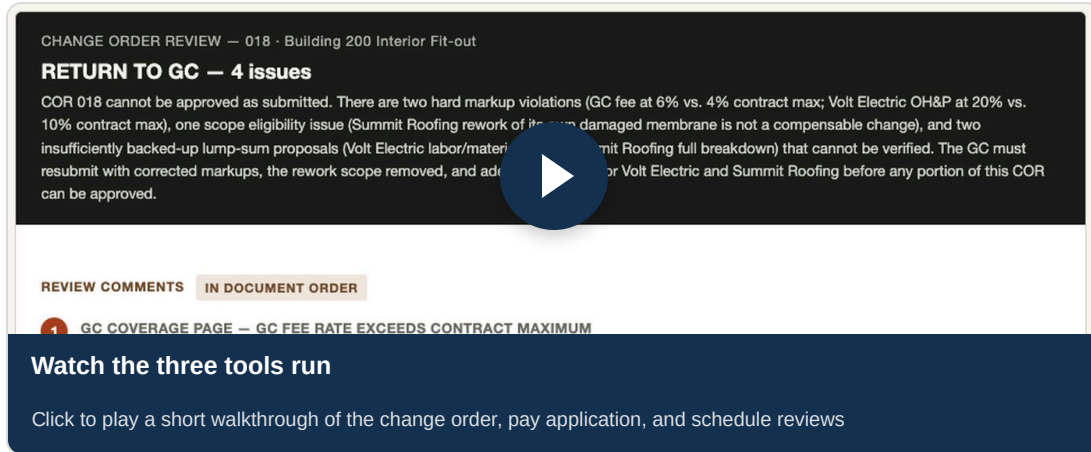
This is the layer AI can work on. It cannot fix the incentives, and it cannot prevent every cause of drift, since site conditions, labor, and outside events move projects even when the information is well managed. What it can do is read across the records one person cannot hold at once and surface the conflicts early enough to act on them. That includes records that disagree with each other, because a submittal log goes out of date and a meeting note can contradict a later instruction. A tool that hides those conflicts creates false confidence. One that exposes them gives the project manager the finding and the evidence behind it.

Risk management is the highest-value work a project manager does. AI's deepest value is making the risk picture visible early, across the fragmented systems where it currently hides.

That is a long way from where most teams are today, and it cannot be built in one step. It starts with single reviews, one document at a time, where the findings can be checked. That is where I started.

Putting the approach into practice

Everything to this point is observation. To test it, I built three working prototypes on my own time for the reviews every project manager repeats. Each uses the same division of work. AI interprets the documents, code checks the defined calculations and rules, and the project manager reviews the findings and decides what to do. That split is how you build for trust.



Change Order Reviewer

Markups, math, and scope that is really rework

Pay Application Reviewer

Schedule of values, retainage, waivers, and non-allowable costs

Schedule Risk Analyzer

Critical path, long-lead risk, and what-if scenarios

Nothing is auto-approved. Each tool produces a review in document order, a ready-to-send note or action plan, and a clear line between what it checked and what stays the project manager's decision.

WHAT THESE ARE, AND ARE NOT

They are prototypes. Single-file browser tools that send a document to a language model, parse the structured response, and run the verification math in plain code. They were tested against documents I constructed, and real documents, with their inconsistent formats and missing backup, will be more difficult. Building them meant working through real failures. The pay application reviewer broke on long documents until I redesigned how it handles them. They demonstrate the approach and the architecture, not a production system. That is the honest state of the work, and closing that gap is the part I most want to do.

WHAT BUILDING THEM TAUGHT ME

The rules matter more than the model. Most of the work was writing down how a reviewer actually checks a document, the markup caps, the backup a package needs, the order a project manager reads it in.

A tool has to say what it could not check. Handed a bare lump sum, the change order tool asks for backup rather than inventing a finding, because a reviewer trusts a tool that admits a gap more than one that guesses.

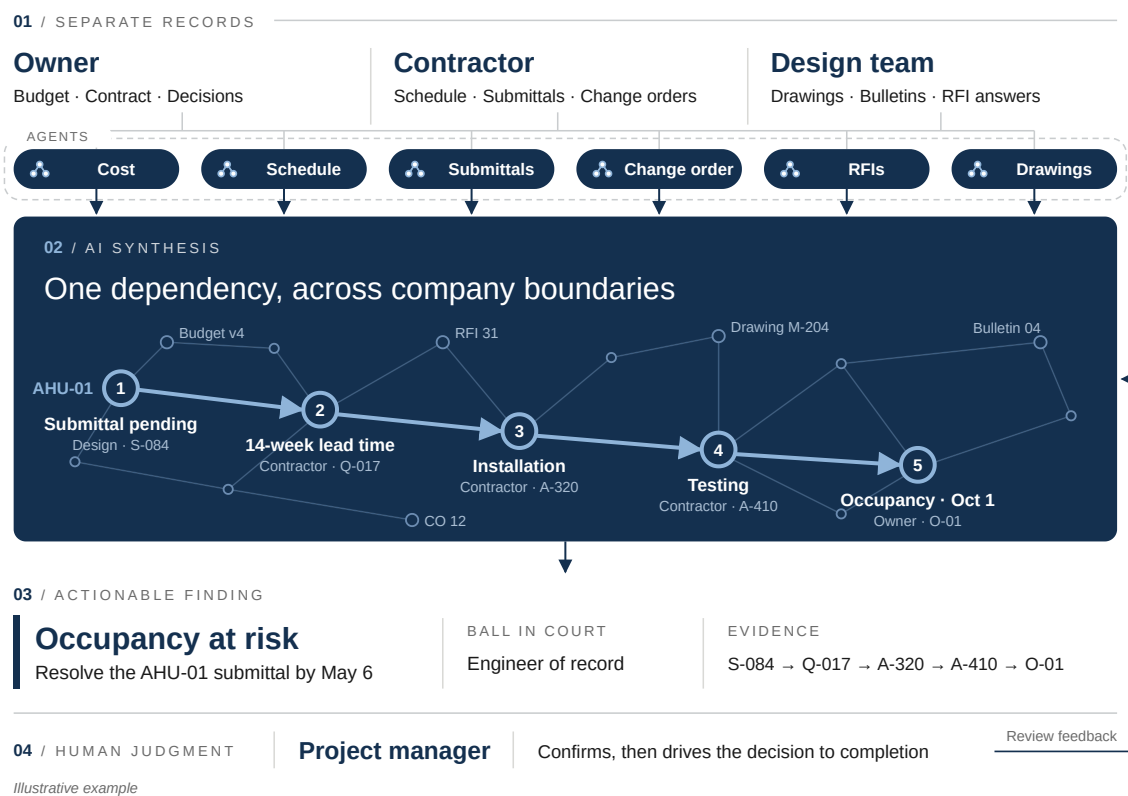
The output has to match how the work is done. A review in document order with a ready-to-send note gets used. A list of anomalies does not.

Change orders and pay applications also record risk that has already materialized. Seeing it earlier depends on running these reviews as agents and connecting what they find, and that is where teams that adopt these tools stand to gain the most.

From single reviews to acting on risk early

Each prototype is one review, run when a person asks for it. The next step is the same review running on its own, and agents like this are now on the market. An agent watches for the document, reads the contract and project records it needs, drafts the review with its sources, and hands it to the project manager to confirm or correct. The corrections are kept, so each review is better than the last.

The larger opportunity is what happens when those agents share what they find. A submittal waiting on approval is routine. The same submittal on equipment with a fourteen-week lead time is a risk to occupancy, and no single review sees it. A layer that reads across every party's records can. Most of these records already reach the owner's project manager as contract deliverables, the schedule updates, submittals, pay applications, and RFI responses. The gap is not access. It is that they arrive as separate documents, and no system reads across them to find where the risk is hiding. McKinsey's 2026 report on the industry describes the same destination from the contractor's side, with agents reading across the model, drawings, procurement, and schedule.



The same applies to a design answer holding up a pour, a change affecting adjacent trades, or missing test evidence threatening turnover. Each finding arrives with its evidence, whose court it is in, and a date to act by, inside the tools the team already uses.

HOW IT GETS BUILT

- **Automate first.** Baseline where the hours go, then put approved tools on the repeatable load.
- **Augment next.** Targeted reviews built with the project managers who use them, then run as agents once trusted.
- **Intelligence after.** Connect the agents into the synthesis layer, one adopted tool at a time.

Adoption is the part most technology rollouts skip, and it is where they fail. Tools built without the practitioner's input get rejected, no matter how good they look in a demo. Building alongside the project managers who will use the tools, starting from what they already have, is how trust forms. Trust is what drives adoption, and adoption is where the return comes from.

Why this is worth investing in now

Every firm will have AI tools. The return goes to the firms that point them at the right work and make someone accountable for making them stick. The value shows up in six places.

CAPACITY

More work, same team

Hours returned across a team add up to people. Growth gets absorbed without hiring in lockstep.

COST

Money caught before it leaves

Reviews catch markups over the cap, unapproved changes billed, and retainage errors before payment. One catch can cover the cost of the tool.

SCHEDULE

Problems found while they are cheap

A risk surfaced weeks early is a decision. Found at the milestone, it is extended general conditions and a delayed opening.

CONSISTENCY

The same standard on every project

Every review runs the same checks, whoever is doing it and however busy they are. Quality stops depending on the individual.

KNOWLEDGE

Experience that stays

Lessons learned, playbooks, and past reviews become something the team can ask. When an experienced project manager leaves, what they knew stays.

PEOPLE

Better work for the people doing it

Project managers spend more of the week on the work they were hired for. New staff get productive faster, and good people are easier to keep.

Together they show up where a client notices it most, fewer surprises, decisions made on time, and a documented reason behind every call.

THE COST OF WAITING

Most firms already pay for AI tools, and most of those tools are underused. Deloitte's latest enterprise survey found that fewer than sixty percent of employees with access to approved AI tools use them in their daily work. The tools are not the constraint. Ownership is, and the cost of missing ownership is paid every month, in hours lost to workarounds, in inconsistency from one project manager to the next, and in the picture a client sees. Meanwhile the foundations take time. Clean records, standard processes, and reviews people trust do not appear overnight, and the synthesis layer depends on all of them. Those capabilities are arriving now. The country's largest contractor has put an AI assistant in every employee's hands, and its people have built hundreds of their own agents. Firms that start now will have the foundations in place. Firms that wait will start from zero, behind competitors who did not.

The return does not come from buying AI. It comes from someone owning the work of making it useful.

WHAT IT TAKES TO START

Not a large program. Someone who owns it, the approved tools a firm already has, a baseline of where the hours go, and a few small trials measured against it. Some tools will be built in house at little cost and others bought, and keeping that spend modest and tied to a clear return is part of the job. Each step has to prove its value before the next. Appendix B sets out how to measure it.

The role pays for itself quickly. On a team of twenty project managers, one person who owns this work covers their own cost by returning two hours a week to each of them. On a larger team the bar is lower still. Everything past that is return, and it compounds every month.

Where project delivery meets AI

I am Ryan Prentice, a Senior Project Manager based in Boston. I have spent fifteen years in construction project delivery as an owner's representative, across higher education, manufacturing, and corporate projects. My perspective comes from working through the coordination, decisions, and relationships that determine how a project actually runs, and I have felt every challenge in this document myself.

I have also always been interested in how the work could run better. When a recurring task takes more effort than it should, or a process makes a decision harder to reach, I want to understand why. For the last three years I have been teaching myself to build with AI, and building has made me more precise about what a tool can do and what still needs a project manager to lead, judge, and decide.

The prototypes are the latest example of a pattern that runs through my career. I see a process that could be improved and I build something to fix it. At one employer I rolled out a task management system that spread on its own as other teams adopted it, and a project plan template the department standardized on. At another I wrote a guideline that became a campus-wide standard. Continuous improvement is the throughline of my career, and AI is the most powerful tool I have found for it.

A recent CliftonStrengths assessment described the same pattern. My top five are Achiever, Learner, Ideation, Futuristic, and Responsibility. Someone who ships rather than plans endlessly, who learns constantly in a field that changes monthly, who sees the connections others miss, who can picture where this is heading and bring people along, and who takes ownership of making the work better. That is not a list I picked to fit. It is how I am wired.

I want the next part of my career to bring project delivery and AI together, shaping how the tools fit the work and building practical ways to make that work better.

For the first time, the tools to make that happen are here. I have never been this clear on what I want to build. I use these tools on my own time because I want to, and where this is heading feels obvious to me. Most project teams have not had the time or the support to think that far ahead. That is the gap I want to close, helping the people who do this work use what is now possible to do it better.

I am looking for a role where I can own this work end to end. If your team is building toward it, or thinking about where to start, I would welcome a conversation and a live walkthrough of the prototypes.

What decides all of this is not what AI can do. It is what a project manager will rely on. That is the standard I build to.

Ryan Prentice

Senior Project Manager · Boston

rprentice99@gmail.com

[LinkedIn](#)

[Prototype walkthrough](#)

Appendix A, the AI opportunity landscape. Appendix B, how to know it is working. Appendix C, how the prototypes work and what a dependable review needs.

The AI opportunity landscape

A working catalog of places AI could support delivery. These are starting judgments about bounded uses, to be tested against the work of a particular team. The same idea can be a simple aid working from supplied records or a substantial integration across live project systems. The place to start is a recurring problem with accessible evidence, a manageable review, and an owner who will support the trial.

Value. High is material recurring time or decision exposure. Medium is a narrower or less frequent benefit.

Effort. Low uses approved tools. Medium needs configuration or a focused build. High needs integration or new controls.

Readiness. Available works with today's tools. Validate needs testing on the task. Research is still uncertain at this scope.

Agents. Anything Available or Validate can eventually run on its own, triggered by a new document or update. The last group lists the agent workflows that matter most.

OPPORTUNITY	VALUE	EFFORT	READINESS
Administration and coordination			
Meeting summaries and proposed follow-up	High	Low	Available
Meeting briefs from supplied records	Medium	Low	Available
Status report drafts from approved inputs	High	Low	Available
Email drafts and thread summaries	Medium	Low	Available
Action log creation and tracking across meetings	High	Low	Available
Presentation drafts from project data	Medium	Low	Available
Document filing across project systems	Medium	High	Validate
Planning and controls			
Cost reconciliation and forecast support	High	Medium	Validate
Cash flow scenarios from approved inputs	High	Medium	Validate
Estimate assumptions and completeness review	High	Medium	Validate
Schedule checks and bounded scenarios	High	Medium	Validate
Automatic baseline schedule from mixed records	High	High	Research
Initial risk register from supplied records	Medium	Low	Available
Budget estimates from historical cost data	High	High	Validate
Design and engineering			
Scope questionnaires and design checklists	Medium	Low	Available
Submittal comparison with stated requirements	High	Medium	Validate
Drawing and specification consistency review	High	High	Validate
Broad constructability judgment across disciplines	High	High	Research

From commercial review to operational readiness

OPPORTUNITY	VALUE	EFFORT	READINESS
Commercial management and procurement			
RFP drafts from approved requirements	Medium	Low	Available
Bid comparisons and possible scope gaps	High	Medium	Validate
Contract clause comparison with a supplied checklist	High	Medium	Validate
Change order review with stated rules and backup	High	Medium	Validate
Pay application reconciliation and document checks	High	Medium	Validate
RFI drafting from supplied context	Medium	Low	Available
Long-lead tracking linked to decisions and schedule	High	High	Validate
Quality, commissioning, and operations			
Inspection and safety record completeness checks	High	Medium	Validate
Punch and closeout document review	High	Medium	Validate
Commissioning prerequisites and test evidence review	High	Medium	Validate
Handover completeness and asset data checks	High	Medium	Validate
Field progress comparison with supported models	High	High	Available
Knowledge, standards, and portfolio insight			
Askable playbook of the processes and workflows on a program	High	Medium	Available
Standard operating procedures drafted from current practice	High	Low	Available
Searchable lessons learned across past projects	High	Medium	Validate
Dashboards for individual projects from live records	High	Medium	Available
Portfolio reporting from structured project records	High	Medium	Available
Cross-system dependency alerts with linked evidence	High	High	Validate
Calibrated prediction of project outcomes	High	High	Research
Agent workflows			
Reviews that run on arrival, change orders, pay applications, submittals	High	Medium	Validate
Inbox agents that file documents, log items, and draft follow-ups	Medium	High	Validate
Schedule watch that re-runs risk analysis on every update	High	High	Research
Meeting agents that prepare the brief, capture the notes, and create the tasks	High	Medium	Validate

How to know it is working

A useful tool should improve something the team can observe. Start by understanding the existing task, including the effort to gather the information, produce the output, and review it. A baseline makes it possible to tell whether a change has helped and whether the benefit lasts after the first demonstration.

MEASURE	WHAT WOULD COUNT AS EVIDENCE
Use in real work	Use per eligible task, repeat use, and reasons for non-use. A monthly review should be assessed against monthly opportunities to use the tool.
Time returned	Total effort from collecting inputs to an accepted output, including checking and correction. Record active work and elapsed turnaround separately.
Quality of the review	Correct findings, unsupported findings, material issues missed, and false alerts against independently checked examples. Check whether the supporting sources are usable.
Earlier decisions	Issues brought forward, decisions made, and action taken before a commitment or milestone. Record the evidence connecting the finding to the response.
Consistency and team experience	Whether reviews and records become more consistent, and whether the tool makes the work easier. Ask users where they still duplicate or repair the output.
Capacity and net value	Work absorbed with quality maintained, allowing for project phase, complexity, and support. Include software, model, maintenance, and review costs.

USING THE RESULTS TO DECIDE WHAT HAPPENS NEXT

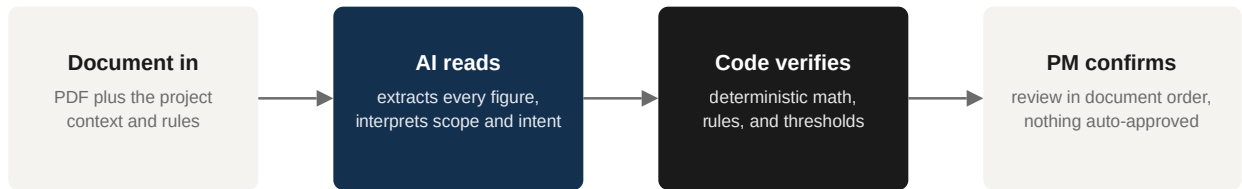
Measures should be small enough to maintain and specific enough to change a decision. A simple record of use, effort, corrections, and outcomes is more useful at the start than a dashboard. As the tools spread, reporting can show those measures by workflow and version, including unresolved failures and the support each tool needs.

A tool can save time and still miss something important, so review quality belongs beside productivity, and the test should reflect the consequence of error. A missed material payment issue deserves different treatment from an awkward sentence in a meeting summary. Claims about avoided losses need particular care, because the alternative outcome cannot be observed. Record what was found, what changed, and the assumptions behind any estimate of value.

Targets get set once the baseline exists, and reviewed like any project. Measured openly, the tools stay accountable, and that is part of how trust gets earned. The case for AI in project delivery is best made the way any good case is made, with data.

How the prototypes work

The three tools share one architecture. The split between what the model does and what the code does is deliberate, and it is the reason a project manager can trust the output.



AI for judgment. Code for arithmetic. The person for the decision.

WHERE THE MODEL ENDS AND THE CODE BEGINS

The model reads the document and the project context and returns a structured record of every figure and every possible finding. Code then recomputes the markups, subtotals, and retainage from those figures against the supplied rules, so the arithmetic in the review is calculated rather than estimated. The schedule analyzer follows the same split, with a critical path engine computing dates from the logic the model extracts. What the code cannot guarantee is that every input was read correctly, which is why the review shows the inputs alongside the result.

THE BOUNDARY OF EACH REVIEW

Each tool also has a line it does not cross. The change order reviewer flags a possible rework issue, but whether the work is entitled needs the original scope and project history. The pay application reviewer checks the documents, but actual completion, material condition, and entitlement to payment need evidence from the field. The schedule analyzer infers relationships the contractor did not mark, and those stay assumptions until the scheduler confirms them.

WHAT A DEPENDABLE REVIEW NEEDS

The next stage makes each finding traceable to a document revision, page, provision, and calculation, so the reviewer can tell a fact from an inference, see what could not be checked, and correct an input without redoing the review.

REQUIREMENT	WHAT TO ESTABLISH
Test cases	Independently checked examples with expected results. Include valid packages, missing backup, conflicting revisions, unreadable pages, and long documents.
Evaluation	Check extraction, support for findings, material misses, false alerts, repeated-run consistency, and reviewer effort. Test whether insufficient evidence produces a useful request for clarification.
Changes over time	Retain versions and test results. Re-run relevant cases when the model, prompts, rules, or document handling change. Set acceptance criteria before judging the result.

MOVING INTO AN ORGANIZATION

A production version needs approved infrastructure, controlled access to project records, audit history, an owner, and a support path. Document content is treated as input, never as authority to change the system's instructions. The handoff includes the architecture, interfaces, rules, test results, and known limits. These are requirements for what comes next, not capabilities the prototypes have established.